

BANK MARKETING CAMPAIGN

EXECUTIVE SUMMARY:

This project analyzed 11,162 customer records from a bank's term deposit marketing campaign to identify what drove subscriptions, and to translate those patterns into concrete recommendations for the next campaign. Data cleaning and exploration were performed in Excel and SQL, with subscription rate breakdowns validated using both COUNTIFS based formulas and native PivotTables.

The analysis found that customers aged 65 and above converted at the highest rate of any age group (80.4%), with students close behind (74.7%). The single strongest predictor of a “yes” was contact history: customers with a successful previous campaign outcome converted at 91.3%, versus 40.7% for customers never previously contacted. Conversely, repeated calling within the same campaign showed diminishing and eventually negative returns conversion fell from 53.4% at one contact to 29.2% at six or more.

Recommendations center on reprioritizing outreach by contact history, capping repeat calls, favoring cellular contact, and shifting campaign weight toward historically stronger months (September–December, March).

Related terms that were researched to know the depth.

What is a Term Deposit? A **Term deposit** is a deposit that a bank or a financial institution offers with a fixed rate (often better than just opening deposit account) in which your money will be returned back at a specific maturity time.

What is credit? In banking, it represents the trust a financial institution places in a customer's ability and willingness to pay them back.

BUSINESS QUESTION/ CONTEXT

How can the financial institution have a greater effectiveness for future marketing campaigns? To answer this, we have to analyze the bank most recent campaign performed to identify patterns that will help inform the next one.

OBJECTIVES

- Understand a business problem from a data perspective.
- Analyze business data to uncover trends and patterns.
- Identify factors influencing business outcomes.
- Communicate findings effectively to stakeholders.
- Make data driven business recommendation

DATASET OVERVIEW

Header	Definition
Rows	11,162
Columns	17
Age	Age of customer - Numerical
Job	Job of customer - categorical
Marital	Marital status of customer categorical
Education	Customer education level
Default	Has credit in default? - categorical
Housing	If costumer has housing loan - categorical
Loan	Has Personal Loan - categorical
Balance	Customer's individual balance
Contact	Communication type - categorical
Month	Last contact month of year - Numerical
Day	Last contact day of the week – Numerical
Duration	Last contact duration, in seconds. - Numerical
Campaign	Number of contacts performed during this campaign and for this client
Pdays	Number of days that passed by after the client was last contacted from a previous campaign. - Numerical
Previous	Number of contacts performed before this campaign and for this client. - Numerical
Poutcome	outcome of the previous marketing campaign. - categorical
Deposit	has the client subscribed a term deposit? - categorical

Data Cleaning

The data was cleaned using SQL and cross-checked using excel tools and functions. Here is the list of cleaning that was done.

- Removed duplicates
SELECT DISTINCT * (Confirmed that there were no blank rows)
- Standardized date formats
- Fixed inconsistent spelling
- Changed data type
- Used TRIM ()
- Checked missing values (Found none)

Exploratory Data Analysis (EDA).

Demographic and behavioral cuts explored: age, job, marital status, education, contact method, prior campaign history, and account balance.

Subscription Rate by Age Group

Age Band	Subscription %
65+	80.4%
18–25	73.4%
26–45	48.5%
46–55	41.2%
56–65	45.7%

Conversion is U-shaped by age: the youngest and oldest customers convert well above the 47.4% overall average; the 26–55 working-age band converts below it, likely reflecting competing financial commitments (mortgages, loans, dependents).

Subscription Rate by Contact Method

Contact Method	Subscription %
Cellular	54.3%
Telephone	50.4%
Unknown	22.6%

Cellular contact substantially outperforms unknown or landline contact. This is as much a data-completeness issue as a channel preference one “unknown” likely reflects missing or unreachable numbers rather than genuine channel rejection.

Analysis (Question Answered)

1. Why do some customers subscribe to term deposits while others do not?

Three factors explain most of the variation:

- Contact history previously reached and previously successful customers convert far above cold contacts.
- Financial cushion - higher balance customers convert more (57.5% in the top balance quartile vs. 36% in the bottom).
- Competing obligations - customers holding a housing loan convert notably less (36.6% vs. 57.0% for those without one).

2. Which customer groups are more likely to respond positively to marketing campaigns?
 - Customers aged 65+ (80.4%) and students (74.7%).
 - Customers with a successful previous campaign outcome (91.3%).
 - Customers in the top account-balance quartile (57.5%).
 - Single customers convert somewhat above married customers (54.3% vs. 43.4%),).

3. Which age group responds best?

Customers aged 65+ generated the highest subscription rate of any age band (80.4%), making them the most responsive segment in this campaign, with students the next strongest group (74.7%).

4. Does campaign frequency matter?

Yes, and the relationship is inverse. Subscription rate declines steadily with repeated contact within the same campaign: from 53.4% at one contact down to 29.2% at six or more. Beyond roughly three attempts, additional calls appear to cost more in customer goodwill and call-center time than they return in conversions.

5. Does previous campaign outcome influence subscriptions?

Substantially. Customers whose previous campaign outcome was a success converted at 91.3% this time the single strongest predictor identified in this analysis, well ahead of any demographic factor.

6. What patterns did you discover?

Customers without a housing loan subscribe at a notably higher rate (57.0% vs. 36.6%).

7. What customer segment requires attention?

Segment	Observation
Retired + high balance	Retired customers carry above-average balances but convert below what balance alone would predict an interaction effect worth a deeper look, e.g. into their investment preferences or risk appetite.
Married, 36–55, holding a loan	Largest population segment in the dataset and one of the weakest converters (40%). Needs a differentiated pitch likely lower disposable income, not lower interest.
Age 65+ × cellular contact	Older customers respond especially well to cellular contact specifically, which may itself be part of why this group converts so highly overall.

Risks

- Calling customers too often reduces engagement conversion nearly halves between 1 and 6+ contacts.
- Heavy reliance on a small number of high converting segments (students, retirees) could limit the campaign's ability to scale.
- Running campaigns in historically weaker months (e.g. May, at 32.8% vs. a 47.4% baseline) risks wasting budget.

Opportunities

- Prior-contact history is the strongest available lever (91.3% success rate) and is likely underused if the current calling list isn't prioritized by it.
- Shifting “unknown method” contacts to cellular could lift response meaningfully once better numbers are sourced.
- Concentrating campaign timing in September–December and March could improve ROI at no added cost.

Recommendation

- Personalize offers based on previous campaign outcomes.
- Allocate a larger share of campaign resources toward customer occupations that demonstrated the highest historical conversion rates (for example, retired customers or students if supported by the data), while testing personalized messaging for lower-performing segments.
- Prioritize channels with better response rates.
- Develop specialized campaigns for customers without housing loans.